Foundations of Data Engineering and Analytics

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Sheet 2: Data Preparation and Data Quality

Exercise 1 (Data Wrangling) [5 Points]

Download the Spotify data set (file `spotify_modified.json`) from OLAT.

a) *1 Point* You will notice that this format is not particularly suited for data analysis. Bring the data into a better format, transform and reshape the data.

b) *1 Point* Investigate whether *danceability_%* contains outliers using at least two different detection methods. Compare the methods and their results. Discuss whether the identified outliers should be considered data quality problems.

c) *2 Points* There are some other anomalies in the dataset. Find them, document your findings, handle them appropriately and describe the methods applied.

d) *1 Point* Handle missing data appropriately. Which method did you choose? Why?

Exercise 2 (Social Data and Biases) [5 Points]

Read the following paper regarding biases in social data, especially Sections 3–8, to learn which biases social datasets may be exposed to:

Olteanu, A., Castillo, C., Diaz, F., & Kıcıman, E. (2019). Social data: Biases, methodological pitfalls, and ethical boundaries. Frontiers in Big Data, 2, 13. doi: 10.3389/fdata.2019.00013

Come up with your own hypothetical *social* study (including human data), including a research question, data collection strategy, etc. Consider this a theoretical case study.
Identify which biases, issues, and challenges your study and dataset may face. For each identified issue, discuss possible mitigation strategies and explain which limitations may remain unresolved.

a) *1 Point* Define your research question, target population, and intended dataset. Explain how you plan to obtain the data and identify general limitations of your proposed study.

b) *1 Point* Are there issues at the *data source level*? How can you mitigate these?

c) *1 Point* Could your *data collection* and *processing* introduce additional biases? If so, which ones and how could you mitigate them?

d) *1 Point* Which types of *analyses* would you choose for your study? Discuss how your methodological choices could introduce or amplify biases and whether combining different approaches could mitigate these risks.

e) *1 Point* How do you plan to *evaluate and interpret* your findings? Which issues may arise, particularly regarding the generalizability of your conclusions?

Important: Submit your solution to OLAT and mark your solved exercises with the provided checkboxes. The deadline ends at 23:59 on the day before the discussion.